

PROJECT PRESENTED BY:

SAFA SABIR 70175350

MALAIKA ABBAS

70177529

Project Name :

Supply Chain Transparency System

➤ Introduction

The Supply Chain Transparency System is designed to provide end-to-end visibility of products from production to delivery. By tracking every batch of goods, the system ensures that quality standards are met, logistics are monitored in real-time, and certifications are verified to maintain trust across the supply chain.

➤ Specific Project Requirements:

- **Batch Tracking:** The system must record unique production batches and link them to specific suppliers and products.
- **Logistics Monitoring:** Real-time tracking of shipments using GPS coordinates and location timestamps.
- **Quality Assurance:** Every batch must undergo an inspection process with detailed results (Pass/Fail) and defect reporting.
- **Compliance Management:** Storage and tracking of certifications (like ISO or Organic) for both suppliers and specific product batches.
- **Personnel Management:** Assignment of staff members to specific roles such as Drivers, Inspectors, or Warehouse Managers.

➤ Identifying entities and attributes:

<u>Entities</u>	<u>Attributes</u>
Suppliers	SupplierID (PK), SupplierName, ContactPerson, Email, Phone, Address, ComplianceStatus
Products	ProductID (PK), ProductName, Category, Description, UnitOfMeasure, BasePrice
Batches	BatchID (PK), ProductID (FK), SupplierID (FK), Quantity, ManufactureDate, ExpiryDate, Status
Tracking	TrackingID (PK), BatchID (FK), CurrentLocation, UpdateTimestamp, Status (In Transit, Received), GPS_Coordinates
Certifications	CertID (PK), BatchID (FK) or SupplierID (FK), CertName (ISO, Organic, etc.), IssuingBody, ExpiryDate, DocURL
Shipments	ShipmentID (PK), BatchID (FK), StaffID (FK), Origin, Destination, DepartureTime, ArrivalTime, CarrierName
Warehouse	WarehouseID (PK), LocationName, Capacity, CurrentOccupancy, ManagerName, Address
Quality Assurance	QA_ID (PK), BatchID (FK), StaffID (FK), InspectionDate, Result (Pass/Fail), DefectDetails, Severity
Staff	StaffID (PK), FullName, Role (Inspector, Driver, Manager), WarehouseID (FK), ContactNumber

➤ **Core System Functionalities**

- Batch Lifecycle Management: The system must track every batch from the ManufactureDate to the ExpiryDate, ensuring no expired goods remain in the supply chain.
- Real-Time Logistics Monitoring: Utilizing the Tracking entity, the system provides live updates on a batch's CurrentLocation and GPS_Coordinates during transit.
- Quality Control Gatekeeping: Every batch must pass a mandatory inspection recorded in the Quality Assurance table before it can be marked as "Received".
- Supplier Compliance Verification: The system checks the ComplianceStatus of a supplier and cross-references it with their Certifications (e.g., ISO, Organic) before allowing them to ship new batches.

➤ **Advanced System Features**

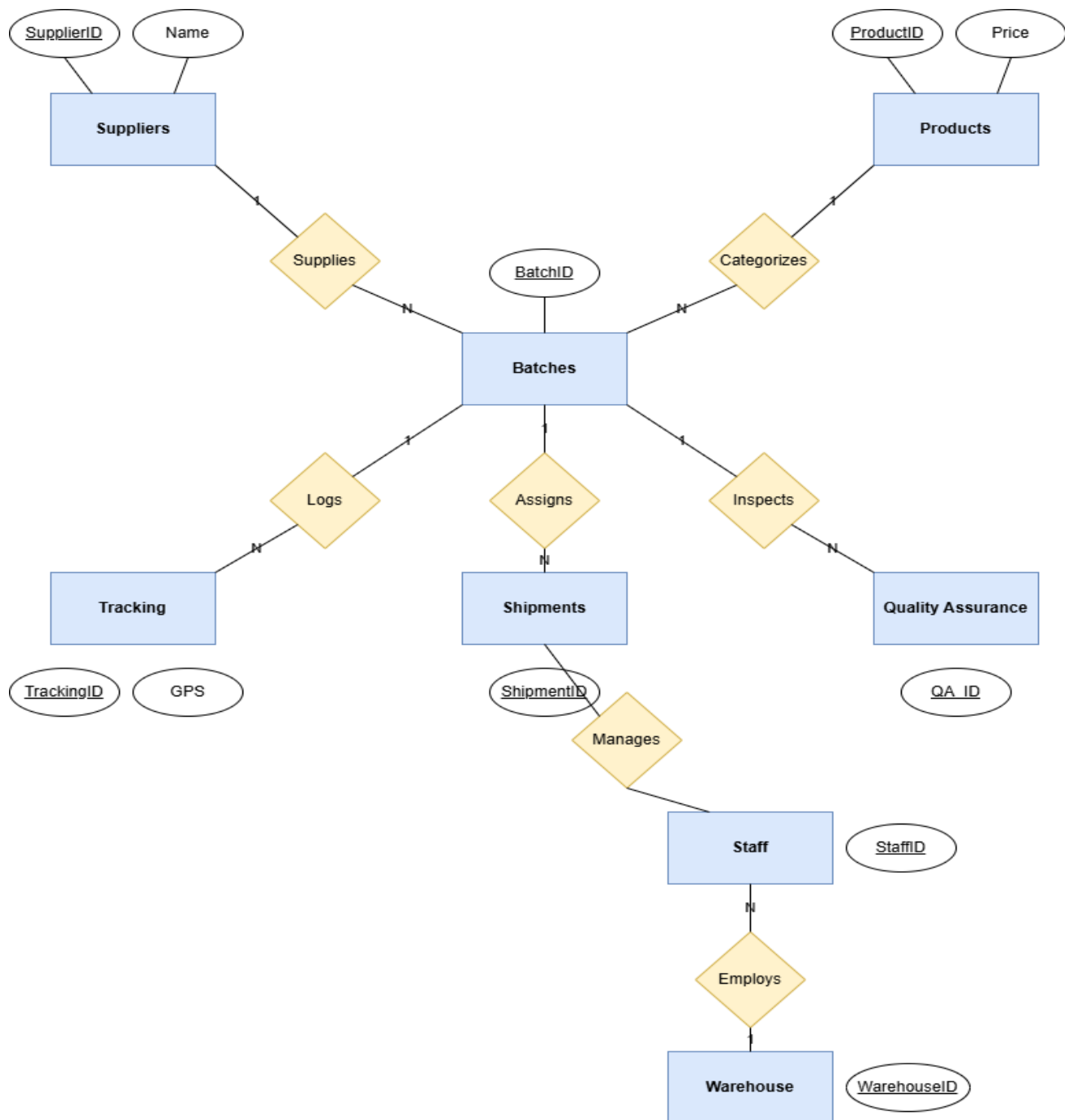
- Warehouse Optimization: A feature to monitor CurrentOccupancy against the total Capacity of each warehouse, helping managers decide where to route incoming shipments.
- Automated Status Transitions: The system automatically updates a shipment's status from "In Transit" to "Received" once the ArrivalTime is logged and the QA Result is confirmed as "Pass".
- Personnel Role Assignment: Functionality to assign specific Staff (Drivers, Inspectors, Managers) to Shipments and Warehouses, ensuring accountability at every step of the chain.
- Digital Documentation Access: A dedicated feature to retrieve DocURL links for certifications, allowing stakeholders to instantly verify the authenticity of products.
- Defect Analytics: A reporting tool that uses DefectDetails and Severity from the QA table to identify recurring issues with specific suppliers or product categories.

➤ **Table for Documentation:**

Table Name	Primary Key	Foreign Keys	Relationship Description
Batches	BatchID	ProductID, SupplierID	Connects products to their specific suppliers.
Tracking	TrackingID	BatchID	Tracks the physical movement of a specific batch.
Shipments	ShipmentID	BatchID, StaffID	Links a batch to a carrier and a responsible staff member.
QA	QA_ID	BatchID, StaffID	Links a batch to an inspector and a pass/fail result.
Staff	StaffID	WarehouseID	Connects employees to their base warehouse.

➤ **ERD (Entity Relationship Diagram):**

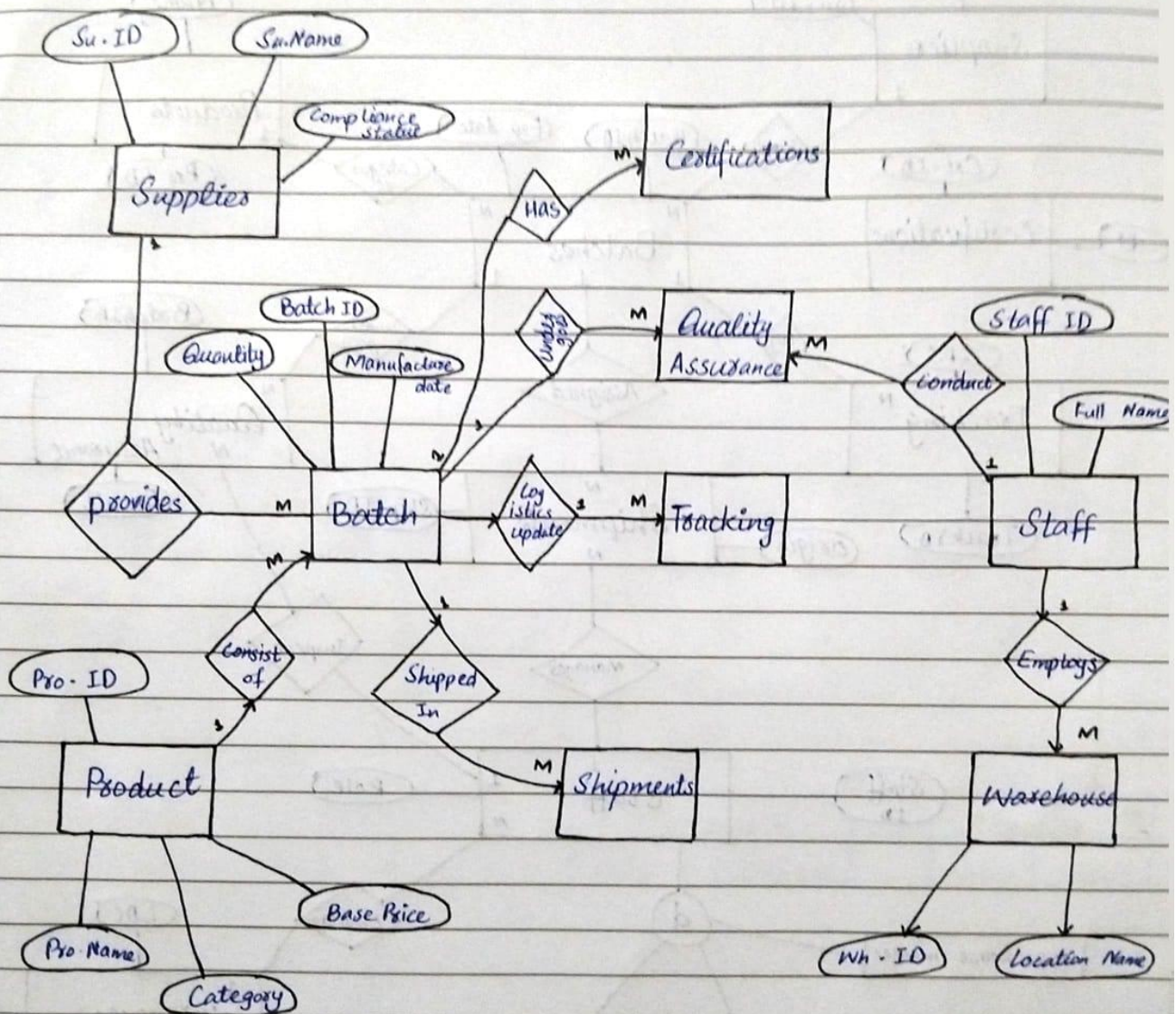
DRAW IO:



Date _____

Mon Tue Wed Thu Fri Sat Sun

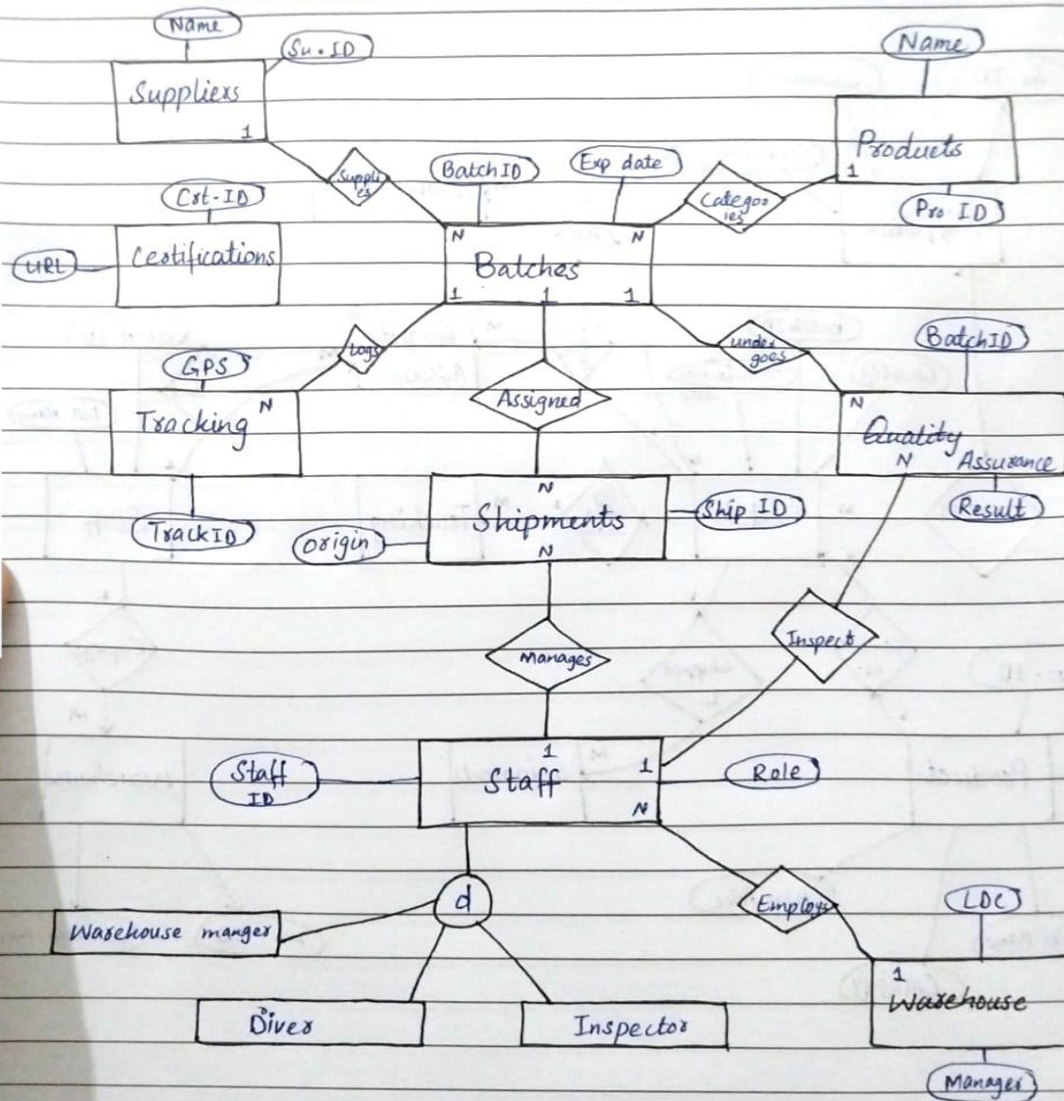
(ERD) Entity Relationship Diagram



ENHANCED ENTITY RELATIONAL DIAGRAM (EERD) :

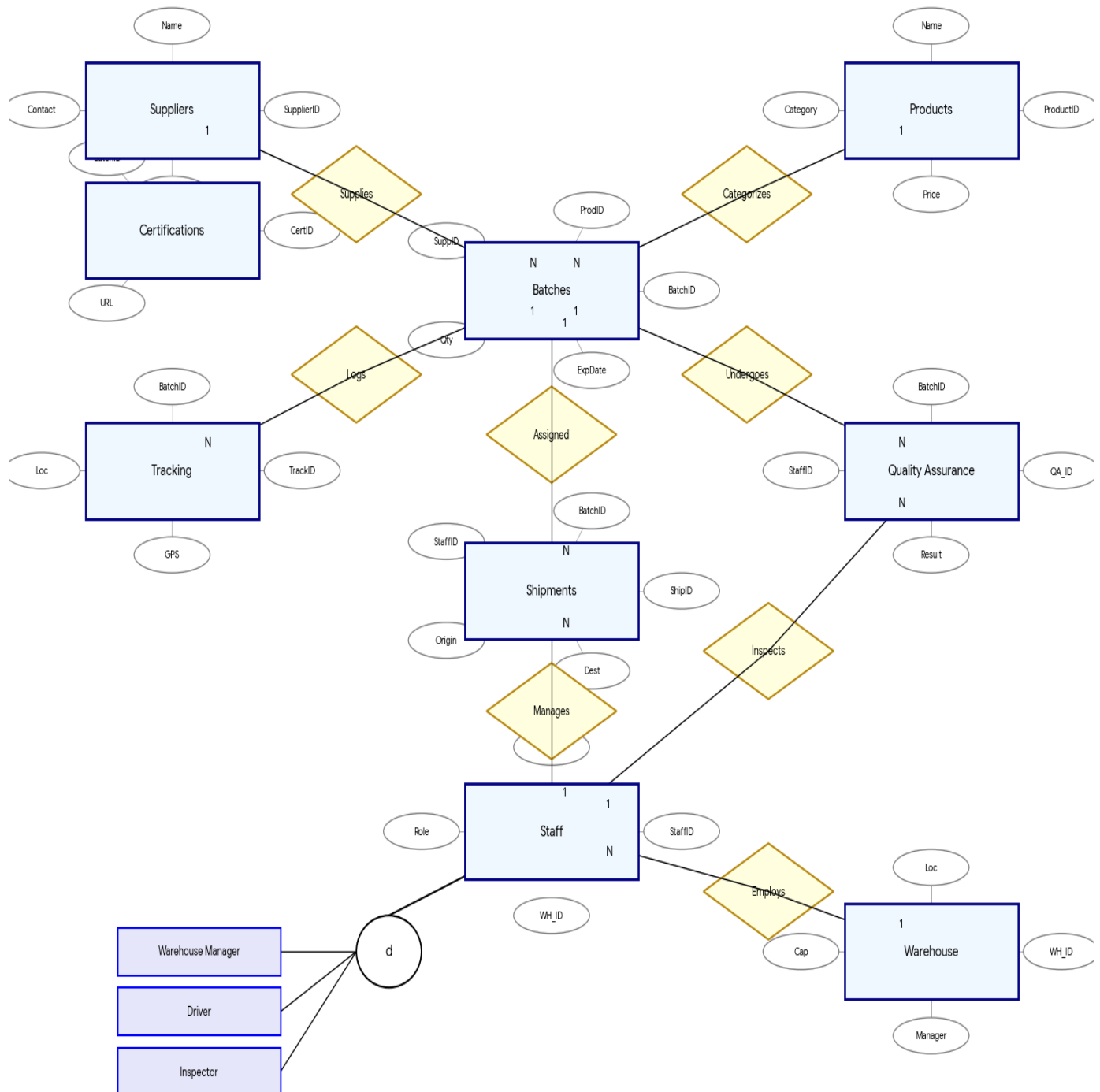
Date _____

Enhanced Entity Relationship Diagram (EERD)



DRAW IO:

Enhanced Entity Relationship Diagram (EERD) Supply Chain Transparency System

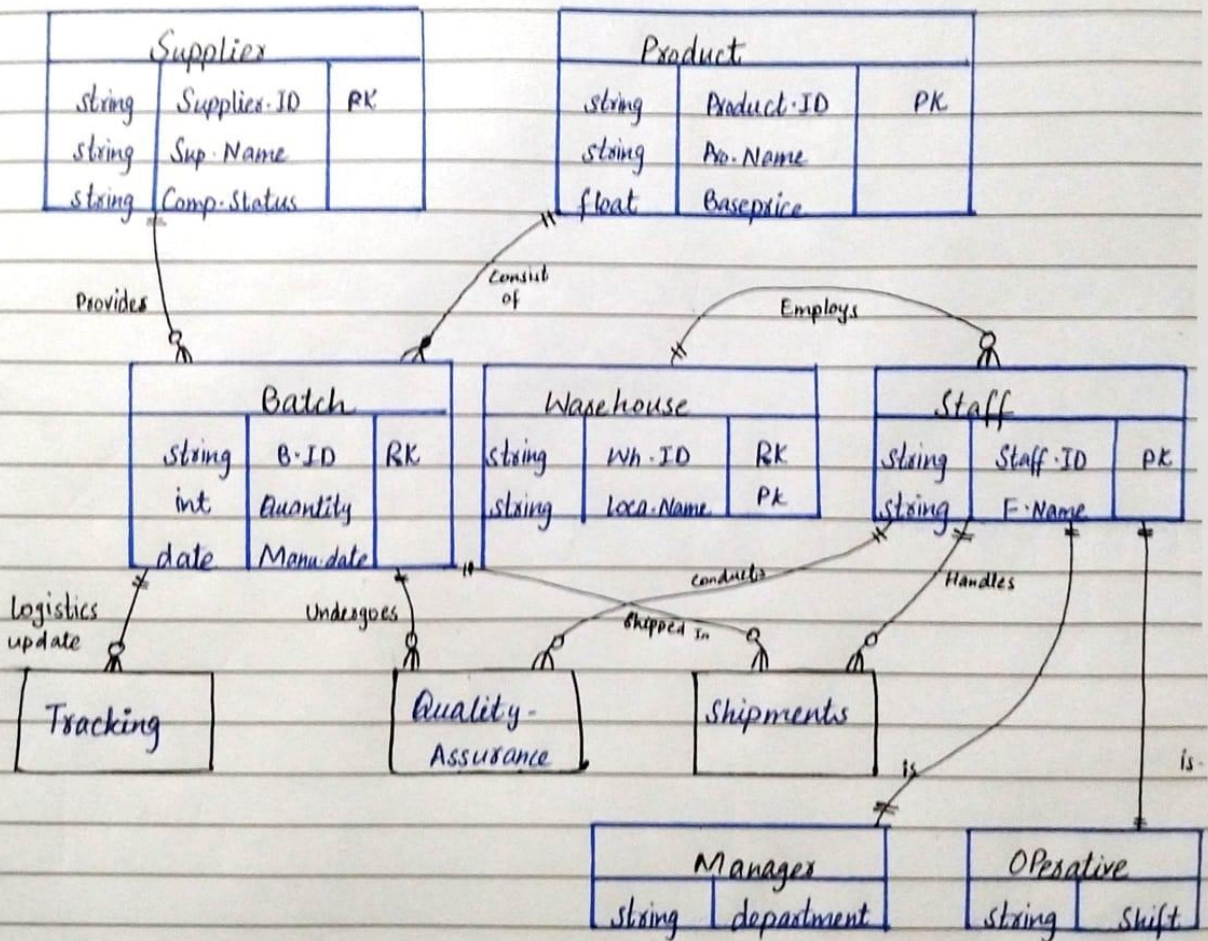


RELATIONAL SCHEMA:

Date _____

Mon Tue Wed Thu Fri Sat Sun

Relational Schema



DRAW IO DIAGRAM:

